

Digital Financial Infrastructure and Financial Inclusion in India

PS 647 Group Project · State-level panel, FY2018-FY2024

HYPOTHESIS

H1. Greater financial infrastructure raises financial inclusion, but its effect on **access** is stronger than its effect on **usage**.

Rationale: opening an account is a supply-side decision banks can be pushed into. Using it is a demand-side decision shaped by income, trust and informality. Infrastructure removes the first constraint, not the second.

Expected: $\beta_{\text{access}} > \beta_{\text{usage}} > 0$, with the difference statistically significant.

VARIABLES

Role	Variable	Indicators (each normalised 0-1)
IV	DFI Index	Branches per lakh adults; ATMs per lakh adults; PMJDY accounts per 1,000 adults
DV 1	Access	Deposit accounts per adult; credit accounts per adult
DV 2	Usage	Deposits per capita; credit outstanding per capita
Derived	AUG	Access – Usage
Control	Economic	log NSDP per capita
Fixed effects	–	State; year

Construction rule. Infrastructure goes on the left-hand side; accounts and balances on the right. Never both, as that produces mechanical correlation. **Weights.** Follow RBI FI-Index logic: Access 35%, Usage 45%.

DATASETS

Source	What it provides	Level	Frequency
RBI Handbook of Statistics on Indian States	Branches, deposits, credit, NSDP per capita, socio-demographics	State	Annual
RBI Quarterly BSR, Statement 3A	State-wise reporting offices, aggregate deposits, bank credit	State	Quarterly
RBI State-wise and Region-wise Deployment of ATMs	ATMs by state and region	State	Annual
RBI BSR-1 / BSR-2 (CIMS-DBIE)	Number of deposit and credit accounts	State	Annual
PMJDY dashboard	PMJDY accounts, RuPay cards issued. Live page is a current snapshot; back years need archived releases	State	Snapshot
Population Projections 2011–2036 (Technical Group)	Adult population denominators for all per-capita ratios	State	Annual

MODEL

Panel of state i and financial year t . FY2018-FY2024, approximately 30 states and UTs, $N \approx 210$. **Two-way fixed effects.**

$$\text{Access}_{it} = \alpha_i + \lambda_t + \beta_1 \cdot \text{DFI}_{it} + \gamma \cdot \ln \text{NSDP}_{it} + \varepsilon_{it} \quad (1)$$

$$\text{Usage}_{it} = \alpha_i + \lambda_t + \beta_1 \cdot \text{DFI}_{it} + \gamma \cdot \ln \text{NSDP}_{it} + \varepsilon_{it} \quad (2)$$

$$\text{AUG}_{it} = \alpha_i + \lambda_t + \beta_1 \cdot \text{DFI}_{it} + \gamma \cdot \ln \text{NSDP}_{it} + \varepsilon_{it} \quad (3)$$

- α_i : state fixed effects. Absorb everything fixed within a state: geography, banking legacy, 2011 baselines.
- λ_t : year fixed effects. Absorb national shocks: UPI growth, COVID, PMJDY drives.
- **Identification:** within-state, over time, relative to the national trend that year.
- **Standard errors:** clustered by state.
- **Equation (3) is not a third finding.** By linearity its β_1 equals $\beta_1^{(1)} - \beta_1^{(2)}$. It supplies the standard error on that difference, which is the formal test of H1.

Diagnostics. (a) Within-state variation in the DFI index. Run this first; if DFI barely moves within states, fixed effects have nothing to estimate from. (b) Hausman test, FE versus RE.

EXPECTED RESULTS

Outcome	Interpretation
$\beta_{\text{access}} > \beta_{\text{usage}} > 0$	H1 supported. Expansion was formal rather than functional.
$\beta_{\text{access}} \approx \beta_{\text{usage}}$	H1 rejected. Genuine deepening of inclusion. Still a finding.
Both ≈ 0	Check within-state variation before concluding anything.
β_1 on AUG > 0 , significant	The access-usage gap <i>widens</i> as infrastructure expands.

LIMITATIONS

- State-level correlation, not person-level. Cannot claim that people with accounts do not use them.
- Deposits are recorded by branch location, not depositor residence, which inflates Maharashtra and Delhi. BSR reports credit by both place of sanction and place of utilisation.
- No connectivity component in the DFI index; reframed as *banking and payments infrastructure*.
- Reverse causality: wealthier states attract more infrastructure. Fixed effects mitigate but do not solve this.